

Climate preconditioning and the mechanics of hazardous alpine mass movements: a catalog and attribution protocol

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Abstract

Large rock, ice, and mixed-material collapses can transform into far-traveled debris flows or locally extreme tsunamis. Warming changes glacier geometry, permafrost temperature, meltwater supply, and precipitation phase, but these changes are seldom separated from immediate weather and non-climatic controls in event reports. I define a catalog and attribution protocol that makes this separation explicit. A convenience seed of eight records—six occurred-event candidates, including two in one site cluster, one dependent episode, and one active-slope case—tests the schema and reconciles source ambiguities. The reviewed seismic record classifies the August 2026 Nepal source as a glacial collapse; the Dickson Fjord rockslide occurred in 2023 rather than the publication year 2024; and Barry Arm remains an active slope rather than an occurred catastrophic failure. The cases require different predictors. The Aru glacier detachments motivate basal-effective-pressure bounds, whereas Chamoli, Dickson Fjord, Tracy Arm, and Barry Arm require glacier-geometry and rock-slope observations in addition to atmospheric reanalysis. I specify climate-blind event discovery, clustered analysis units, detrended seasonal controls, matched non-failing slopes, and multiplicity control. The seed was selected to motivate the study and therefore cannot test a systematic climatic association. No such association is estimated here.

1 Introduction

Alpine mass movements become especially hazardous when an initial failure changes material or enters a confined channel. The 2021 Chamoli rock and ice avalanche transformed into a debris flow and flood that scoured valley walls and struck two hydropower projects [8]. Fjord failures instead transfer momentum to water. The September 2023 Dickson Fjord rockslide generated near-source runup of about 200 m and a very-long-period signal that persisted for nine days [6, 10]. The August 2025 Tracy Arm landslide produced a measured maximum runup of 481 ± 7.6 m [9]. A useful catalog must describe the initial failure separately from the downstream cascade.

The mechanical origin of the initial failure remains the central uncertainty. Warming may remove glacier support from a rock slope, thaw ice in fractures, or change meltwater pressure beneath a glacier. Weather may provide a shorter perturbation through rain, snow loading, melt, or rapid temperature change. Lithology, adverse structure, progressive damage, valley geometry, glacier dynamics, and earthquakes remain competing or interacting explanations. A climatic trend can therefore precondition a failure without being its immediate trigger, and a warm event year does not establish causation.

Here, I use a literature seed to define a reproducible catalog and the tests that would be needed to evaluate systematic association with climate. The seed is a protocol-development set, not an

inference sample. Its purpose is to identify quantities required by the published mechanisms, expose unresolved alternatives, and define how a climate-blind catalog, reanalysis, remote sensing, and matched controls will be combined.

2 Catalog construction and sampling protocol

The catalog consists of three linked tables. The event table records identity, group, analysis role, date, location, setting, initial failure, and downstream transformations. Dates and coordinates have field-specific sources. The measurement table stores one quantity per row and distinguishes a point value, approximation, lower bound, upper limit, or range. It records uncertainty type, observation period, scenario, evidence kind, and source locator separately. The claim table labels each statement as observation, preliminary observation, reconstruction, model output, published inference, or project interpretation. Each event has explicit process-chain, immediate-trigger, and climate-preconditioning claims, including an unresolved claim when evidence is absent.

The draft geophysical inclusion rule requires rapid failure of rock, glacier ice, or a mixed mass in an alpine or fjord setting, a documented time and source location, and at least one of three operational thresholds: initial volume of 10^6 m^3 , source-to-deposit travel of 2 km, or landslide-tsunami field runup of 10 m. Downstream damage and population exposure are stored as consequences but do not determine geophysical inclusion. Snow avalanches without glacier-ice or rock failure and floods without an initial mass movement are outside the scope. The final registration will justify or revise these numerical thresholds before screening. A separately labeled active slope may enter the protocol-development set when measured deformation and published failure geometries exist. Barry Arm is such a prospective case, not an occurred event [1–3, 7].

The present records were selected because they motivated the study. They cannot support a frequency trend, an event-conditioned climate distribution, or a global susceptibility model. Before climate values are extracted, the inference catalog will be preregistered and frozen from a climate-blind search. The registration will state databases, search strings that do not contain a climate cause, language coverage, temporal and spatial bounds, minimum source documentation, duplicate rules, treatment of events detected only by seismic or satellite data, and a stopping date. Two reviewers will screen identity and mechanical eligibility without viewing event climate anomalies. Detection era, sensor coverage, and reporting source will be stored because event detection is not stationary.

The protocol has two estimands. The episode-level trigger estimand compares conditions at failure with time controls; the site-level preconditioning estimand compares source slopes with an incidence-density risk set. The two Aru glaciers are separate episode candidates but share a site-level cluster. The October 2023 Dickson Fjord episode is stored as dependent on the September site and excluded as an independent event-time unit. Barry Arm is excluded from event-time inference because catastrophic failure has not occurred; it instead supplies a prospective case for testing susceptibility variables.

3 Mechanics represented by the seed records

The two Aru glacier tongues detached after surge-like acceleration on low-angle terrain; the published mean Aru-1 surface slope is approximately 13° . Remote sensing and thermomechanical modeling indicate that reduced basal friction shifted resistance toward frozen tongues and lateral margins until those forces were exceeded [4, 5]. Gilbert and colleagues inferred a five- to six-year increase in melt and rain rather than exceptional seasonal forcing. For Aru-1, approximately 10–25 mm of precipitation during the two days before failure is a possible short perturbation; Aru-2 has no

resolved immediate trigger. The two collapses traveled approximately 6 and 5 km beyond their respective glacier termini [5].

At Chamoli, $26.9 \times 10^6 \text{ m}^3$ of rock and glacier ice failed from Ronti Peak and transformed into a highly mobile debris flow [8]. The reported 95% reconstruction interval is $26.5\text{--}27.3 \times 10^6 \text{ m}^3$. The mass was about 80% rock and 20% ice; these are model-dependent composition estimates rather than direct material measurements. Mapping and reconstruction do not require a glacial-lake outburst to explain the flood cascade. Geologic structure and progressive damage remain alternatives to thermal and hydrologic preconditioning. Snow and ice loading on the headwall and a preceding positive temperature anomaly have been proposed as final perturbations, but a unique immediate trigger was not isolated [13].

At Dickson Fjord, a foliation-parallel rockslide struck a gully glacier before entering the fjord. The reconstructed rock volume was $25.5 \pm 1.7 \times 10^6 \text{ m}^3$, with a separate $2.3 \pm 0.15 \times 10^6 \text{ m}^3$ entrained-ice estimate [10]. Rockslide, glacier impact, tsunami, and the very-long-period signal are observations or reconstructions. The cross-fjord seiche was first inferred by wave and seismic modeling; sparse SWOT altimetry later provided direct water-level observations consistent with its phase and period [6]. Multidecadal glacier lowering supports a debuitressing hypothesis, but no proximate trigger or formal anthropogenic event attribution was demonstrated.

Tracy Arm and Barry Arm illustrate occurred and prospective consequences of glacier retreat along fjords. The mapped Tracy Arm volume gives a lower limit of $63.5 \times 10^6 \text{ m}^3$ with a reported uncertainty of $2.7 \times 10^6 \text{ m}^3$; unmeasured submarine failure may increase the total. Several days of accelerating microseismicity preceded failure [9]. Regional summer warming since 1875 was quantitatively attributed to anthropogenic forcing. Observed retreat of South Sawyer Glacier supports debuitressing as a mechanical inference, but the glacier retreat itself was not assigned an anthropogenic counterfactual fraction. At Barry Arm, slope displacement and glacier retreat were temporally coincident during 2010–2017 [3]. Their reported correlation of 0.99 compares cumulative trending series and is not by itself a causal test. Published geometry scenarios span $290\text{--}689 \times 10^6 \text{ m}^3$ [1], while a four-layer seismic inversion estimates a local mean failure-surface depth of $188 \pm 9 \text{ m}$ [2]. These are uncertain slope geometries, not a failed volume.

Table 1: Seven focal records in the protocol seed. The dependent October 2023 Dickson Fjord episode remains in the machine-readable catalog but is omitted here. Volumes describe an initial failed mass except at Barry Arm, where the range describes possible unstable-slope geometries. Reported uncertainties are reproduced from the sources; their distinct confidence, conservative-sum, or unspecified semantics are stored in the machine-readable catalog.

Event and date	Initial failure	Volume (10^6 m^3)	Cascade	Climate evidence
Nepal, 26 Aug 2026	Glacial collapse	unresolved	Debris flow and flood traveled about 100 km	No event-specific attribution; process reconstruction remains provisional [11, 12]

Event and date	Initial failure	Volume (10^6 m^3)	Cascade	Climate evidence
Chamoli, 7 Feb 2021	Rock and glacier ice	26.9 [26.5, 27.3]	Avalanche to debris flow and flood	Several thermal effects are plausible; no single trigger isolated [8, 13]
Aru-1, 17 Jul 2016	Glacier detachment	68 ± 2	Ice avalanche; 6 km beyond terminus	Multiyear melt and rain inferred to increase subglacial water pressure [4]
Aru-2, 21 Sep local time 2016	Glacier detachment	83 ± 2	Two pulses; 5 km beyond terminus	Shared multiyear preconditioning; immediate trigger unresolved [4, 5]
Dickson Fjord, 16 Sep 2023	Foliation-parallel rockslide	25.5 ± 1.7 rock	Glacier entrainment, tsunami, and nine-day signal	Glacier thinning inferred to debutstress the toe; proximate trigger unresolved [10]
Tracy Arm, 10 Aug 2025	Bedrock landslide	≥ 63.5 mapped	Glacier/fjord impact, tsunami, and 36-hour signal	Regional warming attributed; retreat–debuttressing link is a mechanical inference [9]
Barry Arm, active	Slow bedrock landslide	290–689 possible	Potential rapid failure and tsunami	Retreat and motion are temporally coincident; alternatives require testing [1, 3]

4 Mechanical predictors beyond reanalysis

Atmospheric fields do not measure glacier support, permafrost fracture state, or basal effective pressure. The analysis will therefore pair reanalysis with annual optical and radar mapping of terminus position, glacier-slope contact, surface elevation, and source-slope deformation. Define Δh_i as index- date thickness minus baseline thickness at the slope contact. A signed stress-scale proxy for support loss is

$$S_i = -\rho_i g \Delta h_i, \quad (1)$$

where ρ_i is ice density, so thinning gives positive S_i and thickening gives negative S_i . This proxy has stress units but is not the lateral buttressing traction. Geometry-specific elastic or limit-equilibrium calculations are required to translate it into a support change. Results will therefore report signed thickness change, its magnitude, mapped contact loss, and S_i before interpreting debuttressing.

For a first glacier-detachment screening bound, treat the bed as Coulomb till:

$$\begin{aligned} N &= p_i - p_w, & \tau_y &= c + N \tan \phi, \\ \tau_b &\leq \tau_y, & \tau_d &\simeq \rho_i g h \sin \alpha, \end{aligned} \quad (2)$$

where N is effective pressure, p_i is ice overburden pressure, p_w is water pressure, c and ϕ are effective till cohesion and friction angle, h is ice thickness, and α is surface slope. In a slab that neglects lateral and longitudinal resistance, basal yield cannot balance driving stress when

$$N < N_{\text{crit}} = \frac{\tau_d - c}{\tan \phi}. \quad (3)$$

This is a conditional screening range, not an inferred Aru effective pressure; the analysis must span plausible c , ϕ , geometry, and neglected margin stresses. Melt or rain can raise p_w and reduce resistance when drainage is inefficient. The same input can enlarge efficient drainage, lower p_w , and increase resistance.

Permafrost hypotheses require rock-temperature or fracture-ice observations, not air temperature alone. Where boreholes are absent, the protocol will use modeled rock-surface temperature, topographic shading, fracture orientation, and cumulative thermal history as proxies and will label the inference. Adverse structure is a baseline confounder; deformation may be a mediator or competing-process observation. The causal specification will determine which variables are adjusted rather than treating every measured field as a covariate.

5 Event-centered climate test

For each frozen event candidate, reanalysis will provide near-surface air temperature, precipitation phase and amount, snow accumulation, radiative and turbulent fluxes, and surface pressure. Prespecified reductions will include cumulative positive degree days, liquid-water input, recent snow load, freeze-thaw crossings, and a multidecadal temperature trend. Windows, directions, and mechanism-specific primary contrasts will be registered before event values are examined.

Immediate-trigger anomalies and long-term preconditioning require different controls. For trigger variables, I will remove the local seasonal cycle and a smooth long-term trend using data outside the event window. The event residual will be compared with time-matched, same-season blocks. Block resampling will preserve serial correlation, and the two Aru records will share a site-level cluster. This avoids a null that ranks recent event years against an earlier, cooler climate and then interprets the resulting rank as unusual weather.

For persistent preconditioning, the comparison will use incidence-density risk sets. At each event date, eligible controls are catalog-search source slopes that remain under observation and have not yet failed. Controls receive the case index date, and every exposure window ends before that date; a control may later become a case. Follow-up ends at failure, loss of adequate sensor coverage, or the registered study end. Matching uses baseline region, elevation, aspect, relief, lithology where available, and pre-exposure sensor coverage. It does not use index-date glacier contact or deformation, which may lie on the causal pathway. A preregistered causal diagram will distinguish baseline confounders from exposures and mediators.

A hierarchical model will estimate associations while including region and site-group effects, detection era, and monitoring intensity. Glacier-detachment sites will have a signed liquid-water/effective-pressure contrast; glacier-contact rock slopes will have a signed contact-loss/support contrast; and permafrost rock slopes will have a cumulative thermal contrast. Predictors will be standardized against their own risk-set distributions and will not be imputed across inapplicable mechanism classes. Family-wise error will be controlled across these primary contrasts; secondary predictors will use a false-discovery-rate procedure. An association under this design would still not constitute a complete anthropogenic event attribution.

Global products smooth the elevation and aspect that govern steep-terrain energy balance. Downscaling will begin with corrections tied to a named bias: temperature adjusted between grid-cell

and source elevation using a documented lapse rate, precipitation separated by phase, and shortwave radiation corrected for slope, aspect, and shading. Results will be reported before and after each correction. Regional modeling is warranted only when these corrections change an event anomaly enough to affect the prespecified test.

6 Competing hypotheses and discriminating observations

The analysis will evaluate signs and timing that distinguish mechanisms. Debuttressing predicts a deformation change point after glacier thinning or contact loss, with a geometry-dependent lag. Loading has no general sign. For an added load Δq on a planar slope, the simple components are $\Delta\tau = \Delta q \sin \alpha$ and $\Delta\sigma_n = \Delta q \cos \alpha$; the Coulomb tendency $\Delta\tau - \Delta\sigma_n \tan \phi$ changes sign with slope and friction angle, before crack pressure and three-dimensional geometry are considered. Hydrologic weakening predicts velocity or failure probability increasing with water input when drainage is inefficient; hydrologic strengthening predicts a negative relation or seasonal hysteresis as drainage evolves. Permafrost degradation predicts deformation associated with cumulative rock-temperature history and ice-bearing fractures rather than a single warm day. Progressive damage predicts accelerating deformation or microseismicity without requiring unusual weather. Structural release predicts failure along mapped adverse discontinuities. Seismic forcing predicts temporal coincidence with an earthquake capable of producing materially large shaking at the source.

Barry Arm provides a prospective case for checking several predictions. Change points and lags in slope velocity, glacier-contact loss, precipitation, groundwater, and seasonal temperature can be tested with autocorrelated errors. If a glacier term loses explanatory power after detrending or after internal creep state is included, the cumulative 0.99 correlation should not be treated as evidence for a dominant causal pathway.

7 Uncertainty and next steps

Source uncertainty enters at three levels. First, reported intervals are not interchangeable. Chamoli gives a 95% reconstruction interval; Aru reports a conservative linear sum of DEM and post-event correction errors; some Dickson and Tracy values still require verification and are labeled **reported_unspecified**. The Barry Arm depth spread describes agreement among tested parameterizations and time periods, not full inversion nonuniqueness. Second, source classifications can change. The reviewed Nepal seismic record now calls the source a glacial collapse and debris flow, whereas the dated event page retains broader provisional wording [11, 12]. Both records are preserved. Third, some quantities are conditional scenarios. Barry Arm volumes and tsunami heights describe possible geometries, not observations or forecasts of a failure time.

The next version must complete climate-blind discovery and double screening before any event anomalies are calculated. It must then acquire matched non-failing slopes, glacier-contact histories, deformation records, and permafrost proxies. Only after those data and the statistical specification are frozen can the study estimate whether recent hazardous alpine mass movements occupy systematically different climatic or cryospheric conditions.

8 Conclusions

The seed catalog establishes a traceable distinction among initial failure, downstream cascade, climate preconditioning, and immediate trigger. It also shows why reanalysis alone is insufficient: Aru requires effective-pressure bounds, while the rock-slope cases require glacier geometry, support

change, structure, and deformation. A climate-blind inventory, independent analysis units, detrended seasonal controls, matched non-failing slopes, and explicit competing predictions are prerequisites for a defensible climate test. The present contribution is the protocol and machine-readable seed, not evidence that warming systematically triggered these events.

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